

★ n-Ary Trees - An n-ary tree is a tree in which each node has at most n children.

Ex: Binary Tree $\rightarrow$ At most 2 children per node

3-ary Tree $\rightarrow$ At most 3 children per node

k-ary Tree $\rightarrow$ At most k children per node

Strict n-Ary Tree - It is one in which every node can have 0 children or either exactly n children.

If height given $\left\{ \begin{array}{l} \text{Min}^m. \text{ no. of nodes of k-ary Tree} = kh + 1 \\ \text{Max}^m. \text{ no. of nodes of k-ary Tree} = \frac{k^{h+1} - 1}{k - 1} \end{array} \right.$ where h = height

If no. of nodes already given $\left\{ \begin{array}{l} \text{Min}^m. \text{ Height} = \log_k [n(k-1) + 1] - 1 \\ \text{Max}^m. \text{ Height} = \frac{n-1}{k-1} \end{array} \right.$

$$e = (k-1)i + 1$$

external
nodes

internal
nodes

k-ary Tree, value of k.